

Natural language statement

Domain

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Theorem 7.6 (Inverse Function Theorem). Suppose U and V are open subsets of $\mathbb{R}^n$, and $F: U \rightarrow V$ is a smooth map. If $DF(p)$ is nonsingular at some point p in U , then there exist connected neighborhoods U_0 subset U of p and V_0 subset V of $F(p)$ such that $F|_{U_0}: U_0 \rightarrow V_0$ is a diffeomorphism.

Smooth manifolds

```
structure SmoothDiffeomorphismOn {m n : N} (U : Set ((Fin m -> R)))
  (V : Set ((Fin n -> R))) where
  toFun : (Fin m -> R) -> (Fin n -> R)
  invFun : (Fin n -> R) -> (Fin m -> R)
  mapsTo : MapsTo toFun U V
  invMapsTo : MapsTo invFun V U
  leftInvOn : Set.LeftInvOn invFun toFun U
  rightInvOn : Set.RightInvOn invFun toFun V
  smooth : ContDiffOn R top toFun U
  smooth_inv : ContDiffOn R top invFun V

theorem lee_7_6_inverse_function_theorem
  {n : N} {U V : Set ((Fin n -> R))) {F : (Fin n -> R) -> (Fin n -> R)}
  {p : (Fin n -> R)} (hU : IsOpen U) (hV : IsOpen V)
  (hF : MapsTo F U V /\ ContDiffOn R top F U) (hp : p in U)
  (hD : Function.Bijective (fderiv R F p)) :
  exists U0 V0,
    (IsOpen U0 /\ IsConnected U0 /\ p in U0 /\ U0 subset U) /\
    (IsOpen V0 /\ IsConnected V0 /\ F p in V0 /\ V0 subset V) /\
    exists e : SmoothDiffeomorphismOn U0 V0, e.toFun = F := by
  sorry
```

Theorem 7.8 (Rank Theorem). Suppose U subset $\mathbb{R}^m$ and V subset $\mathbb{R}^n$ are open sets and $F: U \rightarrow V$ is a smooth map with constant rank k . For any p in U , there exist smooth coordinate charts centered at p and $F(p)$, with U_0 subset U and $F(U_0)$ subset V_0 subset V , such that $\psi \circ F \circ \phi^{-1}(x_1, \dots, x_k, x^{k+1}, \dots, x^m) = (x_1, \dots, x_k, 0, \dots, 0)$.

Smooth manifolds

```
structure SmoothDiffeomorphismOn {m n : N} (U : Set ((Fin m -> R)))
  (V : Set ((Fin n -> R))) where
  toFun : (Fin m -> R) -> (Fin n -> R)
  invFun : (Fin n -> R) -> (Fin m -> R)
  mapsTo : MapsTo toFun U V
  invMapsTo : MapsTo invFun V U
  leftInvOn : Set.LeftInvOn invFun toFun U
  rightInvOn : Set.RightInvOn invFun toFun V
  smooth : ContDiffOn R top toFun U
  smooth_inv : ContDiffOn R top invFun V

def EuclideanConstantRank {m n : N} (U : Set ((Fin m -> R)))
  (F : (Fin m -> R) -> (Fin n -> R)) (k : N) : Prop :=
  forall x in U, Module.fderiv R (LinearMap.range (fderiv R F
    x).toLinearMap) = k

theorem lee_7_8_rank_theorem
  {m n k : N} {U : Set ((Fin m -> R))) {V : Set ((Fin n -> R))}
  {F : (Fin m -> R) -> (Fin n -> R)} {p : (Fin m -> R)}
  (hU : IsOpen U) (hV : IsOpen V) (hF : MapsTo F U V /\ ContDiffOn R top
    F U)
  (hp : p in U)
  (hk : k <= m /\ k <= n) (hrank : EuclideanConstantRank U F k) :
  exists U0 V0, (IsOpen U0 /\ IsConnected U0 /\ p in U0 /\ U0 subset U) /\
    (IsOpen V0 /\ IsConnected V0 /\ F p in V0 /\ V0 subset V) /\
    MapsTo F U0 V0 /\
    exists (sourceTarget : Set (Fin m -> R)) (targetTarget : Set (Fin n
      -> R))
      (phi : SmoothDiffeomorphismOn U0 sourceTarget)
      (psi : SmoothDiffeomorphismOn V0 targetTarget),
    forall x in phi.toFun '' U0, psi.toFun (F (phi.invFun x)) =
      fun i => if h : i.1 < k /\ i.1 < m then x <i.1, h.2> else 0 := by
  sorry
```

Theorem 7.13 (Rank Theorem for Manifolds). Suppose M and N are smooth manifolds of dimensions m and n , respectively, and $F: M \rightarrow N$ is a smooth map with constant rank k . For each p in M there exist smooth coordinates $(x_1, \dots, x^m)$ centered at p and $(v_1, \dots, v_n)$ centered at $F(p)$ in which F has the coordinate representation $(x_1, \dots, x^m) \Rightarrow (x_1, \dots, x_k, 0, \dots, 0)$.

Smooth manifolds

```
def ConstantRank {m n : N} {M : Type u} {N : Type v}
  [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M]
  [TopologicalSpace N] [ChartedSpace ((Fin n -> R)) N]
  (F : M -> N) (k : N) : Prop :=
  forall p : M,
    Module.fderiv R
      (LinearMap.range
        (mfderiv I(R, (Fin m -> R)) I(R, (Fin n -> R)) F p).toLinearMap) = k

theorem lee_7_13_rank_theorem_for_manifolds
  {m n k : N} {M : Type u} {N : Type v}
  [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M]
  [IsManifold I(R, (Fin m -> R)) inf M]
  [TopologicalSpace N] [ChartedSpace ((Fin n -> R)) N]
  [IsManifold I(R, (Fin n -> R)) inf N]
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		<pre> {F : M -> N} (hF : ContMDiff I(R, (Fin m -> R)) I(R, (Fin n -> R)) inf F) (hk : k <= m /\ k <= n) (hrank : ConstantRank (m := m) (n := n) F k) : forall p, exists (phi : OpenPartialHomeomorph M (Fin m -> R)) (psi : OpenPartialHomeomorph N (Fin n -> R)), phi in IsManifold.maximalAtlas I(R, (Fin m -> R)) inf M /\ psi in IsManifold.maximalAtlas I(R, (Fin n -> R)) inf N /\ p in phi.source /\ F p in psi.source /\ MapsTo F phi.source psi.source /\ forall x in phi.target, psi (F (phi.symm x)) = fun i => if h : i.1 < k /\ i.1 < m then x <i.1, h.2> else 0 := by sorry </pre>
Theorem 8.8 (Constant-Rank Level Set Theorem). Let M and N be smooth manifolds, and let $\ell : M \rightarrow N$ be a smooth map with constant rank equal to k. Each level set of ℓ is a closed embedded submanifold of codimension k in M.	Smooth manifolds	<pre> def ConstantRank {m n : N} {M : Type u} {N : Type v} [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M] [TopologicalSpace N] [ChartedSpace ((Fin n -> R)) N] (F : M -> N) (k : N) : Prop := forall p : M, Module.finrank R (LinearMap.range (mFderiv I(R, (Fin m -> R)) I(R, (Fin n -> R)) F p).toLinearMap) = k def EmbeddedSubmanifoldOfCodimension {m : N} {M : Type u} [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M] (S : Set M) (codim : N) : Prop := forall p in S, exists phi : OpenPartialHomeomorph M ((Fin m -> R)), phi in IsManifold.maximalAtlas I(R, (Fin m -> R)) inf M /\ p in phi.source /\ phi '' (S inter phi.source) = {x in phi.target forall i : Fin m, m - codim <= i.1 -> x i = 0} theorem lee_8_8_constant_rank_level_set_theorem {m n k : N} {M : Type u} {N : Type v} [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M] [IsManifold I(R, (Fin m -> R)) inf M] [TopologicalSpace N] [ChartedSpace ((Fin n -> R)) N] [IsManifold I(R, (Fin n -> R)) inf N] { : M -> N} (h : ContMDiff I(R, (Fin m -> R)) I(R, (Fin n -> R)) inf) (hrank : ConstantRank (m := m) (n := n) k) : forall c, IsClosed {p p = c} /\ EmbeddedSubmanifoldOfCodimension (m := m) {p p = c} k := by sorry </pre>
Corollary 8.10 (Regular Level Set Theorem). Every regular level set of a smooth map is a closed embedded submanifold whose codimension is equal to the dimension of the range.	Smooth manifolds	<pre> def RegularValue {m n : N} {M : Type u} {N : Type v} [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M] [TopologicalSpace N] [ChartedSpace ((Fin n -> R)) N] (F : M -> N) (c : N) : Prop := forall p, F p = c -> Manifold.IsSubmersionAt I(R, (Fin m -> R)) I(R, (Fin n -> R)) inf F p def EmbeddedSubmanifoldOfCodimension {m : N} {M : Type u} [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M] (S : Set M) (codim : N) : Prop := forall p in S, exists phi : OpenPartialHomeomorph M ((Fin m -> R)), phi in IsManifold.maximalAtlas I(R, (Fin m -> R)) inf M /\ p in phi.source /\ phi '' (S inter phi.source) = {x in phi.target forall i : Fin m, m - codim <= i.1 -> x i = 0} theorem lee_8_10_regular_level_set_theorem {m n : N} {M : Type u} {N : Type v} [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M] [IsManifold I(R, (Fin m -> R)) inf M] [TopologicalSpace N] [ChartedSpace ((Fin n -> R)) N] [IsManifold I(R, (Fin n -> R)) inf N] { : M -> N} {c : N} (h : ContMDiff I(R, (Fin m -> R)) I(R, (Fin n -> R)) inf) (hc : RegularValue (m := m) (n := n) c) : IsClosed {p p = c} /\ </pre>

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		<pre> EmbeddedSubmanifoldOfCodimension (m := m) {p p = c} n := by sorry </pre>
Theorem 9.16 (Quotient Manifold Theorem). Suppose a Lie group G acts smoothly, freely, and properly on a smooth manifold M. Then the orbit space M/G is a topological manifold of dimension equal to $\dim M - \dim G$, and has a unique smooth structure with the property that the quotient map $\pi: M \rightarrow M/G$ is a smooth submersion.	Smooth manifolds	<pre> def SmoothFreeProperAction {g m : N} {G : Type u} {M : Type v} [Group G] [TopologicalSpace G] [ChartedSpace ((Fin g -> R)) G] [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M] (act : G -> M -> M) : Prop := (forall x, act 1 x = x) /\ (forall a b x, act (a * b) x = act a (act b x)) /\ ContMDiff (I(R, (Fin g -> R)).prod I(R, (Fin m -> R))) I(R, (Fin m -> R)) inf (fun p : G x M => act p.1 p.2) /\ (forall a x, act a x = x -> a = 1) /\ IsProperMap fun p : G x M => (act p.1 p.2, p.2) theorem lee_9_16_quotient_manifold_theorem {g m : N} {G : Type u} {M : Type v} [Group G] [TopologicalSpace G] [ChartedSpace ((Fin g -> R)) G] [LieGroup I(R, (Fin g -> R)) inf G] [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M] [IsManifold I(R, (Fin m -> R)) inf M] {act : G -> M -> M} (haction : SmoothFreeProperAction (g := g) (m := m) act) : exists (Q : Type v) (_ : TopologicalSpace Q) (_ : ChartedSpace (Fin (m - g) -> R) Q) (_ : IsManifold I(R, (Fin (m - g) -> R)) inf Q) (pi : M -> Q), Surjective pi /\ (forall x y, pi x = pi y <=> exists a, act a x = y) /\ Manifold.IsSubmersion I(R, (Fin m -> R)) I(R, (Fin (m - g) -> R)) inf pi /\ forall (Q' : Type v) (_ : TopologicalSpace Q') (_ : ChartedSpace (Fin (m - g) -> R) Q') (_ : IsManifold I(R, (Fin (m - g) -> R)) inf Q') (pi' : M -> Q'), Surjective pi' -> (forall x y, pi' x = pi' y <=> exists a, act a x = y) -> Manifold.IsSubmersion I(R, (Fin m -> R)) I(R, (Fin (m - g) -> R)) inf pi' -> exists e : Diffeomorph I(R, (Fin (m - g) -> R)) I(R, (Fin (m - g) -> R)) Q Q' inf, e comp pi = pi' := by sorry </pre>
Theorem 10.7 (Sard's Theorem). If $F: M \rightarrow N$ is any smooth map, the set of critical values of F has measure zero in N.	Smooth manifolds	<pre> theorem lee_10_7_sards_theorem {m n : N} {M : Type u} {N : Type v} [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M] [IsManifold I(R, (Fin m -> R)) inf M] [TopologicalSpace N] [ChartedSpace ((Fin n -> R)) N] [IsManifold I(R, (Fin n -> R)) inf N] {F : M -> N} (hF : ContMDiff I(R, (Fin m -> R)) I(R, (Fin n -> R)) inf F) : let critical := {p : M not Manifold.IsSubmersionAt I(R, (Fin m -> R)) I(R, (Fin n -> R)) inf F p} forall psi : OpenPartialHomeomorph N (Fin n -> R), psi in IsManifold.maximalAtlas I(R, (Fin n -> R)) inf N -> volume (psi '' (F '' critical inter psi.source)) = 0 := by sorry </pre>
Theorem 10.11 (Whitney Embedding Theorem). Every smooth n-manifold admits a proper smooth embedding into $\mathbb{R}^{2n+1}$.	Smooth manifolds	<pre> theorem lee_10_11_whitney_embedding_theorem {m : N} {M : Type u} [TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M] [IsManifold I(R, (Fin m -> R)) inf M] [T2Space M] [SigmaCompactSpace M] : exists F : M -> (Fin (2 * m + 1) -> R), IsProperMap F /\ IsEmbedding F /\ ContMDiff I(R, (Fin m -> R)) I(R, (Fin (2 * m + 1) -> R)) inf F := by sorry </pre>
Theorem 10.16 (Whitney Approximation Theorem). Let M be a	Smooth	<pre> theorem lee_10_16_whitney_approximation_theorem </pre>

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smooth manifold and let $F: M \rightarrow \mathbb{R}^k$ be a continuous function. Given any positive continuous function $\delta: M \rightarrow \mathbb{R}$, there exists a smooth function $F': M \rightarrow \mathbb{R}^k$ that is δ -close to F . If F is smooth on a closed subset A subset M , then F' can be chosen to be equal to F on A .

manifolds

```
{m k : N} {M : Type u} [TopologicalSpace M] [ChartedSpace ((Fin m ->
R)) M]
[IsManifold I(R, (Fin m -> R)) inf M] [T2Space M] [SigmaCompactSpace M]
{F : M -> (Fin k -> R)} {delta : M -> R} {A : Set M}
(hF : Continuous F) (hdelta : Continuous delta) (hdeltapos : forall x,
0 < delta x)
(hA : IsClosed A) (hFsmoothOnA : exists U in nhdds A,
ContMDiffOn I(R, (Fin m -> R)) I(R, (Fin k -> R)) inf F U) :
exists Fsmooth : M -> (Fin k -> R),
ContMDiff I(R, (Fin m -> R)) I(R, (Fin k -> R)) inf Fsmooth /\
(forall x, dist (Fsmooth x) (F x) < delta x) /\ EqOn Fsmooth F A :=
by
sorry
```

Theorem 10.19 (Tubular Neighborhood Theorem). Every embedded submanifold of $\mathbb{R}^n$ has a tubular neighborhood.

Smooth manifolds

```
def EmbeddedSubmanifoldOfCodimension {m : N} {M : Type u}
[TopologicalSpace M] [ChartedSpace ((Fin m -> R)) M]
(S : Set M) (codim : N) : Prop :=
forall p in S,
exists phi : OpenPartialHomeomorph M ((Fin m -> R)),
phi in IsManifold.maximalAtlas I(R, (Fin m -> R)) inf M /\ p in
phi.source /\
phi '' (S inter phi.source) =
{x in phi.target | forall i : Fin m, m - codim <= i.1 -> x i = 0}

def IsNormalVector {n : N} (M : Set ((Fin n -> R)))
(x v : (Fin n -> R)) : Prop :=
x in M /\ forall : R -> (Fin n -> R), forall velocity : (Fin n -> R),
0 = x -> (forall t in nhds 0, t in M) -> HasDerivAt velocity 0 ->
sum i, v i * velocity i = 0

def NormalDiskBundle {n : N} (M : Set ((Fin n -> R)))
(radius : M -> R) : Set (M x (Fin n -> R)) :=
{p | IsNormalVector M p.1 p.2 /\ ||p.2|| < radius p.1}

theorem lee_10_19_tubular_neighborhood_theorem
{n codim : N} {M : Set ((Fin n -> R))}
(hM : EmbeddedSubmanifoldOfCodimension (m := n) M codim) :
exists (radius : M -> R) (U : Set (Fin n -> R)),
(forall x, 0 < radius x) /\ IsOpen U /\ M subset U /\
exists inverse : (Fin n -> R) -> M x (Fin n -> R),
Set.BijOn (fun p : M x (Fin n -> R) => (p.1 : (Fin n -> R)) + p.2)
(NormalDiskBundle M radius) U /\
ContinuousOn inverse U /\
ContDiffOn R top (fun z => ((inverse z).1 : (Fin n -> R))) U /\
ContDiffOn R top (fun z => (inverse z).2) U /\
forall p in NormalDiskBundle M radius, inverse ((p.1 : (Fin n ->
R)) + p.2) = p := by
sorry
```